

Aerosol generator

AGK

2000



Operating Manual
For specialized users

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1 About this manual

1.1 Function of the manual

This manual provides you with information about the aerosol generator *AGK 2000*.

Among other things, you will find information on:

- Safety
- Functionality
- Use
- Operation
- Maintenance

1.2 Target group for the manual

This manual is intended for **specialized users** trained at Palas GmbH who are familiar with activities in the field of particle measuring technology based on their technical education or experience.

1.3 Validity of the manual

This manual is valid for the aerosol generator *AGK 2000*.

1.4 Storage of documents

The storage of documents is the responsibility of the owner and/or user to ensure the documents are available when they are needed. Store the documents together with the device.

1.5 Presentation rules

The following presentations are used in this manual:

Text	Product names and product descriptions Example: <i>Fidas</i>
	Cross-references to other documents Example: <i>Information about ... is found in the manual for the ...</i>
Text	Menu items Example: Shut Down System
	Selection and settings Example: Start measuring with Record
Text > Text	Menu paths. The sequence of the menus is indicated by the ">" character. Example: Menu > Check for Updates
„Text“	Word compositions and figurative language Example: "Measuring Unit"

[>32]	Reference to page number Example: For further information, see the section "Technical data [>12]"
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2 For your safety

2.1 Hazards and safety instructions

Qualified person

1. Only a properly qualified person may perform installation, commissioning and any other work on the device.
2. Follow the generally accepted codes of practice when installing the device.

Safe operation of the device

1. Only operate the device with the insulation and/or casing fully installed.
2. Make sure that the housing and/or the casing on the device is undamaged, complete, and correctly installed.

Safe use of the device

1. Do **not** use the device in a flammable or explosive environment, e.g. rich in hydrogen or oxygen.
2. Do **not** use the device with flammable or explosive carrier gases, e.g. rich in hydrogen or oxygen.
3. Please only use dry, particle- and oil-free compressed air supply.
4. Strong acids (e.g. hydrochloric acid, sulphuric acid, nitric acid) as well as aromatic or chlorinated hydrocarbons must not be used as aerosol substance or as solvents.
5. Only use original spare parts! Please contact the manufacturer whenever necessary.
6. The AGK 2000 has been set by the manufacturer to be used with a maximum pre-pressure of $P_{\max}$ 6 bar for the compressed air supply. Please make sure that the maximum pre-pressure of $P_{\max}$ 6 bar specified on the type plate agrees with the provided compressed air line where the machine is to be used.
7. For safety reasons, under no circumstances may the machine be commissioned if there is visible damage. In this event, please contact Palas (Tel.: 0721/96 213-0).

Safety instructions on device

1. Never remove or cover the safety stickers on the device. The stickers have to remain legible for the entire service life of the device.
2. If they become damaged or illegible, replace the safety stickers immediately.

Environment

1. The machine may only be operated in dry rooms, under atmospheric ambient pressure and at room temperature.
2. Operating under other ambient conditions such as in corrosive or explosive environments, in strong electric or electromagnetic fields, in areas with ionising radiation as well as in areas subject to impact and vibration is not covered by any functional guarantee by the manufacturer.

Aerosols

Aerosols might be dangerous to your health. That's why they ought not to be inhaled. It might also be necessary to wear protective clothes. Please pay attention to the correspondent standards and safety rules.

2.2 Warnings

The warnings in this manual are highlighted with pictograms and signal words. The severity of a hazard is indicated by the pictogram and signal word.

Layout of the warnings

The warnings that precede each action are shown as follows:

	DANGER
	Type and source of the hazard
	Explanation of the type and source of the hazard/consequences ► Measures to avert the hazard

Meaning of the signal words

DANGER	Immediate danger to life or risk of severe physical injury if this hazard is not avoided.
WARNING	Possible risk of severe physical injury if this hazard is not avoided.
CAUTION	Possible risk of minor physical injury if this hazard is not avoided.
NOTICE	Damage to property if this hazard is not avoided.

Meaning of the pictograms

	DANGER Warnings with information about the severity of the hazard
	DANGER Danger to life due to electric shock
	NOTICE Damage to property

2.2.1 Warnings on the device

There is a warning sticker on the inside of the device.

Meaning of the pictograms

	Disconnect the power plug before opening the housing.
	Read the operating manual prior to commissioning.

2.3 Regulations

Observe the following regulations and directives:

Legal requirements

- Legal regulations for Health and Safety at Work
- Legal regulations for environmental protection
- Employer's Liability Insurance Association provisions

Standards and directives

- The applicable safety requirements according to DIN, EN, and VDE

2.4 Obligations of the user

To ensure the proper functioning of the device, observe the following:

1. Carefully read the instructions before using the device.
2. Keep the manuals in the vicinity of the device to ensure that information can be checked at all times.
3. Only carry out activities described in the manual intended for you.
4. Make sure that the required inspections and maintenance tasks are carried out.
5. Have the device maintained by the manufacturer.
6. Have damage to the device repaired promptly by the manufacturer.

Maintenance

Repair

3 Product description

3.1 Use

3.1.1 Intended use

The device is manufactured and tested according to the state of the art and recognized safety rules. In order to avoid hazards to yourself or third parties and damage to the device and other assets, only use the device properly and as intended.

The aerosol generator *AGK 2000* was developed in conformance with VDI guideline 3491, part 5, generation of test aerosols by nebulization of dye-solutions with nozzle nebulizers, VDI handbook "Reinhaltung der Luft", volume 4.

The aerosol generator *AGK 2000* was designed to disperse a solution, e.g. NaCL or KCl or a pure liquid into air/ gas stream.

The device is intended for use in indoor areas and laboratories.

The device may be used outdoors in roofed areas protected against the elements.

The manufacturer assumes no liability for damage caused by exposure to the elements.

The device is not intended for use by the following persons:

- Persons with limited physical, sensory, or mental abilities
- Persons lacking experience or knowledge
- Children under 16 years of age

These persons must be supervised by a person responsible for their safety or previously instructed in the use of the device.

Children must be supervised to prevent them from playing with the device.

Any use other than the intended use is prohibited. The manufacturer assumes no liability for damages resulting from this. All warranty claims are voided if changes are made to the device, also in the course of assembly and installation.

All corresponding documentation must be observed for all work on the device.

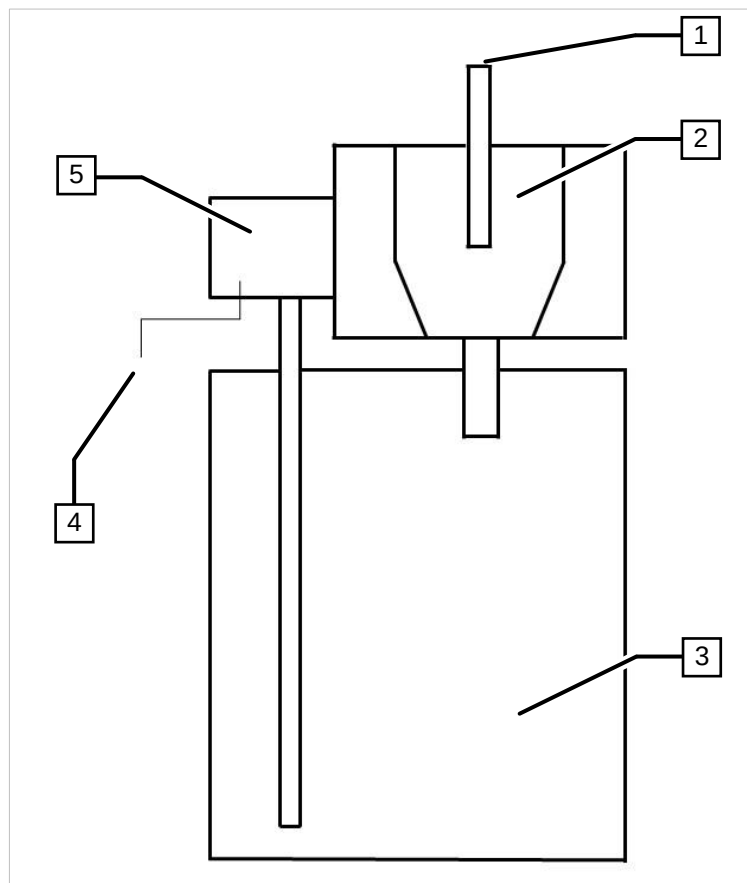
In case of improper actions, the manufacturer is not liable for resulting damages.

3.1.2 Improper use

The manufacturer or supplier assumes no liability for damages due to improper use.

3.2 Functional description

The aerosol generator AGK 2000 nebulizes liquids and potassium chloride solutions (KCl) by means of a binary atomizer nozzle. The following figure shows a schematic drawing of the generator components.



Schematic drawing

1	Aerosol	2	Cyclone
3	Liquid reservoir	4	Pressurized air
5	Nozzle		

Pressurized air is led to the binary atomizer nozzle via an adjustable pressure regulator. The pre-pressure can be adjusted between 0 and 4 bar above ambient pressure and is displayed on the integral pressure gauge. A filter for suspended particles installed in front of the nozzle retains particles that might be in the air. Due to the under-pressure created in the nozzle the liquid is drawn from the liquid reservoir. Due to under-pressure inside the nozzle, the liquid is sucked out of the liquid reservoir.

The aerosol cloud generated in the nozzle flows tangentially into a cyclone. Large droplets are trapped due to centrifugal forces and flow back into the liquid reservoir. The remaining droplets leave the cyclone via the outlet tube. The size distribution of these droplets depends slightly on the nozzle-generated primary droplet distribution but much more on the separation characteristics of the cyclone. In consequence, for many substances the size distribution is independent of the mass flow rate of the liquid in the nozzle.

The separation characteristics of the cyclone depends on the air volume flow rate and as a result it depends on the pre-pressure of the binary atomizer nozzle. With an increasing volume flow rate or pre-pressure the separation limit is shifted to smaller particle diameters.

In general the *AGK 2000* disperses pure liquids, solvents and suspensions. Whereas aerosols from pure liquids of low vapour pressure hardly show any changes while leaving the cyclone, the volatile components of solvents and suspensions evaporate.

When using KCl suspensions, the size of the remaining particles of solutions can be varied by the mixing ratio of the aerosol substance (e.g. KCl) in a volatile solvent (e.g. H₂O).

3.2.1 System description

The *AGK 2000* generator is fitted into a rig that is made of anodised aluminum. The front view shows the pressure control valve with the pressure gauge as well as the pressurized air connection. The telescopic tube of the cyclone separator (outside diameter 30 mm) serves as aerosol outlet. By adding a tube, the aerosol can be transported to the place it is supposed to be used at.

3.3 Type plate

The type plate is on the front of the device.

3.4 Dimensions

H x W x Ø : 400 x 250 x 180 mm

3.5 Scope of delivery

- Inspect the delivered equipment for damage and completeness.

The aerosol generator *AGK 2000* consist of binary nozzle in combination with a cyclone, as well as liquid reservoir. Beside those components the following parts are included:

The scope of delivery includes:

- 1 binary nozzle
- 1 cyclone
- 1 liquid reservoir
- 1 pressure minimizer with manometer

Further:

- 1 manual
- 1 cover of liquid reservoir
- 1 pressure connector adapter NG8



AGK 2000

3.6 Environmental conditions

Environmental conditions affect the functionality of the device.

The system can only be operated in dry rooms, under atmospheric working pressure and at room temperature.

The manufacturer bears no responsibility for charge resulting from operation under different ambient conditions, e.g. in corrosive or explosive environments, in strong electric or electromagnetic fields, in areas with ionising radiation as well as areas with high levels of shock or vibrations.

Operation

The following environmental conditions apply for the operation of the device:

- Temperature range: 10 to 30 °C
- Humidity: ≤ 85%

Storage / transport

The following environmental conditions apply for the storage and transport of the device:

- Temperature range: 0 to 40 °C
- Humidity: ≤ 95%

Operation and storage under other environmental conditions, for instance in corrosive or explosive environments, powerful electrical or electromagnetic fields, areas with ionizing radiation or areas subject to shocks and vibrations is prohibited.

3.7 Optional accessories

The following accessories are available:

- Compressed air supply adapter
- Transport case
- Connector for MFP series test rigs
- *TR 2000* drying stage for the drying of moistness in the aerosol



Both *TR 2000* are connected to the *AGK 2000* by simple plug on top of the *AGK 2000* outlet and with a y-adapter you connect the second volume flow of the *TR 2000*. This y-adapter has to be placed in the tubing line between filter and nozzle. Usual airflow relation between *TR 2000* and *AGK*-nozzle is 2:1 (please see separate page of "Relation between volume air flow rate and pre-pressure", which will be delivery with *TR 2000*).

4 Operation and settings

1. It is essential to read the operating instructions thoroughly before putting the generator into operation.
2. Please check right after the unpacking if the device shows visible transport damages.

4.1 Unpackaging and setup Note

Note

Please take care about the transport weight of the box. Please determine the balance point of the instrument and lift it up carefully for the transport to the final destination.

To unpack the *AGK 2000*, observe the following:

1. Remove the instrument from the transport box.
2. Existing filling material and transport fixation have to be removed.
3. The details on the shipping note have to be compared with the delivered goods.
4. Take care, that at the final destination of the instrument the area is plane and all sufficient space is allowed to make operating elements freely accessible.

4.2 Connection to Pressurized Air Supply

Note

The manufacturer is not liable for damages caused by poor quality of compressed air or by connection to a compressed air supply at a pressure too high.

The aerosol generator *AGK 2000* is manufactured for connection to a compressed air supply with a maximum pressure of 7 bar (absolute). Observe the following:

- In order to take full advantage of the operating range of the generator the minimum pressure at the air supply should be 4 bar (absolute).
- The pressure indicated at the pressure gauge should not exceed 3.5 bar to prevent damage of the pressure gauge.
- The pressurized air should be dry, oil-free and particle-free.
- The connection should be made via a coupling type NG 8.

4.3 Set Up, Short Description

1. Turn pressure minimizer anti-clockwise in order to set the nozzle pre-pressure down to zero.
2. Unscrew liquid reservoir.
3. Fill reservoir with liquid to be atomized.
4. When changing the liquid, the instrument must be cleaned and dried carefully.
5. Screw in liquid reservoir in again.
6. Connect pressurized air to generator.
7. Adjust desired nozzle pre-pressure with pressure regulator.

4.4 Filling of Liquid Reservoir

Note

The reservoir should not be filled up higher than to two third of its height. The quantity of liquid sucked from the nozzle slightly depends on the suction height and thus on the liquid level. If very high stability and reproducibility of the particle mass flow rate are requested, it is recommendable to regularly refill the reservoir up to the indicated level (e.g. every 4 h). Solvents should be completely renewed from time to time as the solution may become concentrated due to solvent evaporation from droplets caught in the cyclone.

Observe the following:

1. The generator must be disconnected from compressed air and/or the actual pressure should be set to zero at the pressure gauge by turning anti clockwise.
2. The liquid reservoir (contents 500 ml) can be taken away by turning it anti-clockwise.
3. Check the correct position of the rubber seal on the bottom of the cyclone before putting the reservoir back in place.
4. The reservoir is then inserted and screwed in place manually.

4.5 Installation of AGK 2000

- The generator outlet simply needs to be connected to the aerosol input location by means of a tube (inner diameter of 30 mm).

If the aerosol must be distributed to different input locations, the volume flow can be divided by means of Y-shaped-tube-connectors.

- In order to achieve a homogenous distribution, the individual tubes should have the same length. The particle size which is generated is not much influenced by transportation through the tube.

Nevertheless, the tubes should be as short as possible.

4.6 Aerosol Generation

	WARNING
	Health risk due to aerosols Aerosols might be dangerous to your health. That's why they ought not to be inhaled. It might also be necessary to wear protective clothes. Please pay attention to the corresponding standards and safety rules.

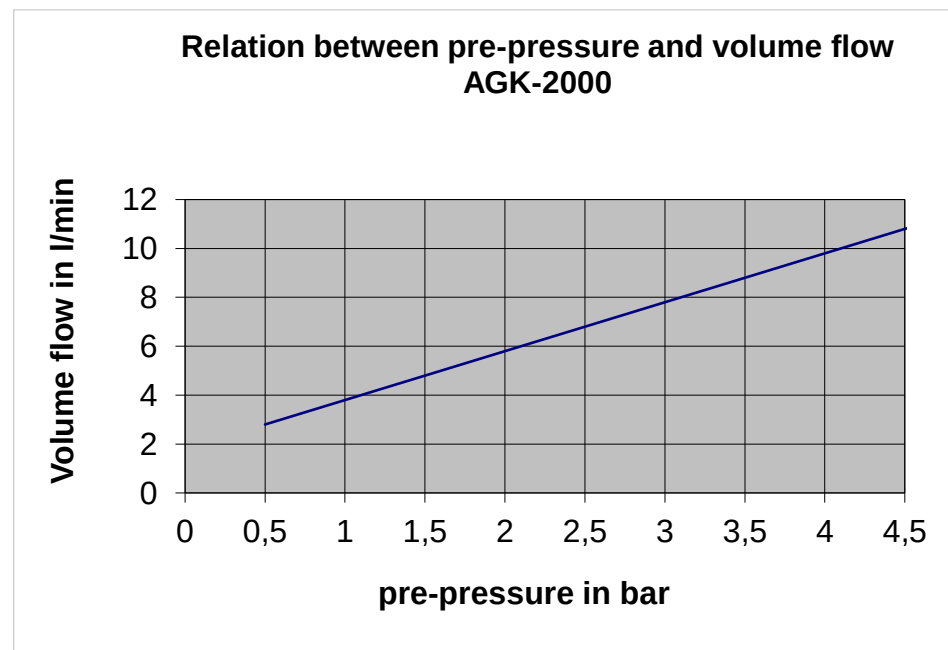
- ▶ Aerosol generation is started by setting the pre-pressure at the pressure control valve. The pressure gauge serves as control unit for the operation level.

After long periods of inactivity or after having refilled the reservoir, the liquid inlet tube may contain air bubbles. It may take some time until the generator produces a steady output.

Nozzle pre-pressure

1. At first we suggest to set the nozzle pre-pressure to about 3.0 bar.
2. As soon as a homogenous aerosol cloud is visible at the tube outlet (dark background makes the cloud best visible) the nozzle pre-pressure can be set to the desired values.
3. To ensure steady generation the pre-pressure should be set to a minimum value of 0.7 bar.

The nozzle pre-pressure influences the generated volume flow as well as the particle production rate (and thereby the produced mass flow). If KCl solutions are used, the mass flow is also influenced by the KCl concentration. The following figure shows the volume flow as a function of the nozzle pre-pressure.



Relation between pre-pressure and volume flow AGK 2000

4.7 Aerosol substances

The aerosol generator *AGK 2000* has been developed especially for the generation of KCl-aerosols out of a KCl solution. Principally, the generator is also suitable for the dispersion of pure liquids and suspensions. Thereby, the user has to pay attention to the nozzle being cleaned carefully with a suitable solvent when the liquid is changed. Otherwise the nozzle could be blocked.

Exchange of liquids

When changing from a watery liquid to an oily one, observe the following:

1. The generator must be cleaned carefully with distilled water in order to remove all salt residues.
2. The generator must be dried.
⇒ Then, it may be run with the oily aerosol substance.

If the change of aerosol substances shall be done from an oily liquid to a watery one, observe the following:

1. The generator has to be cleaned carefully with a suitable solvent first.
2. The user has to pay attention to the generator being totally dry and degreased.

4.8 Working with KCl-solutions

Note

The generator should not be run with a NaCl-solution of more than 10% concentration for a time period longer than 8 h). This could evoke that the nozzle gets blocked with crystalized KCl particles.

After every operation with the generator it must be cleaned with distilled water:

- This is done by running the generator at a pre-pressure of 4 bar for a short period of time in order to clean the atomizer nozzle.

When running the generator with a watery KCl solution, this liquid is dispersed in very fine droplets by the atomizer nozzle. Due to the use of dry pressurized air, a great part of water in each droplet is evaporated already during the process of generation. The remaining water evaporates inside the cyclone separator, so that the aerosol leaving the generator may be called a dry aerosol.

Particle size distribution

The following table shows the impact of the KCl-concentration on the produced particle size distribution (qualitatively). A lower KCl-concentration means shifting the particle size distribution towards smaller particle diameters.

- Particle number distribution (dlogDp)
- KCl-solution (20 weight %)
- 4 bar pre-pressure
- measurement time: 1 min.
- measurement device: Palas PCS-2000

D [μm]	dN	$dN / \sum dN$	q0 (Dp)	Q0 (Dp)
0.255	1286	0.0366	0.5850	0.0366
0.294	4726	0.1344	2.1499	0.1709
0.340	5038	0.1432	2.2918	0.3142
0.392	4620	0.1314	2.1017	0.4455
0.453	3742	0.1064	1.7023	0.5519
0.523	2972	0.0845	1.3520	0.6364
0.604	2308	0.0656	1.0499	0.7020
0.698	1905	0.0542	0.8666	0.7562
0.806	1465	0.0417	0.6664	0.7979
0.931	1303	0.0370	0.5927	0.8349
1.075	1121	0.0319	0.5100	0.8668
1.241	935	0.0266	0.4253	0.8934
1.433	782	0.0222	0.3557	0.9156
1.655	645	0.0183	0.2934	0.9339
1.911	550	0.0156	0.2502	0.9496
2.207	457	0.0130	0.2079	0.9626
2.548	371	0.0105	0.1688	0.9731
2.943	280	0.0080	0.1274	0.9811
3.398	225	0.0064	0.1024	0.9875
3.924	150	0.0043	0.0682	0.9917
4.532	120	0.0034	0.0546	0.9951
5.233	61	0.0017	0.0277	0.9969
6.043	47	0.0013	0.0214	0.9982
6.978	26	0.0007	0.0118	0.9989
8.058	21	0.0006	0.0096	0.9995
9.306	7	0.0002	0.0032	0.9997
10.746	6	0.0002	0.0027	0.9999
12.409	3	0.0001	0.0014	1.0000
14.330	0	0.0000	0.0000	1.0000
16.548	0	0.0000	0.0000	1.0000

Influence of KCl concentration on particle size distribution

D [μm]	Mean diameter	q0	Distribution density (number)
dN	Number of particles	Q0	Number (sum)

4.9 Cleaning of the generator

Note

Please work carefully so that the plastic surface is not scratched or damaged.

- ▶ A suitable solvent together with a soft cloth or soft brush should be used. Please ensure that the tangential aerosol inlet bore is not clogged with fluff from cloth or paper.

Note

If solutions and suspensions are used that contain substances solid at room temperature (e.g. latex particles) or that might crystallize out (e. g. KCl-solutions), the nozzle might become clogged. The generator should therefore be operated with distilled water (with a pure solvent respectively) after every operation, in order to wash out all remaining aerosol substances from the nozzle. If this shall be done, the liquid reservoir must first be emptied and then be re-filled with the solvent (distilled water respectively).

Cleaning of the cyclone

The cyclone can be disassembled for cleaning purposes. Observe the following:

- ▶ After loosening the screws of the cyclone cap with integrated outlet tube, it can be taken apart and the inner part of the cyclone can be cleaned.

Please work carefully so that the plastic surface is not scratched or damaged.

- ▶ A suitable solvent together with a soft cloth or soft brush should be used. Please ensure that the tangential aerosol inlet bore is not clogged with fluff from cloth or paper.

The generator is reassembled in reverse order.

Salt suspensions

After using the instrument with salt suspensions, e.g. NaCl or KCl, the instrument must be cleaned. Observe the following:

1. Exchange the liquid reservoir.
2. Flash the instrument with distilled water.
 - ⇒ Within the distilled water the salt is flashed out of the tubing and nozzle.
3. Let the generator operate approx. 10 or 15 minutes at a pre-pressure of at least 2 bar.

If this procedure was forgotten, it is possible to dismount the dispersion nozzle and clean it within an ultra sonic bath. Observe the following:

- ▶ Therefore the four screws at the nozzle have to be removed and the dispersion nozzle has to be taken carefully out by pulling.

If you reassemble the dispersion nozzle please take care that the part with the opening is facing in the direction of the liquid tubing.

5 Maintenance

Regular maintenance of the device extends the service life and improves operational reliability.

Under normal operation conditions the *AGK 2000* is almost maintenance-free. The filter placed in front of the nozzle should be exchanged if the volume flow rate of the generator clearly decreases below the nominal value.

The cyclone can be disassembled for cleaning purposes. After loosening the screws of the cyclone cap with integrated outlet tube, it can be taken apart and the inner part of the cyclone can be cleaned. Please work carefully so that the plastic surface is not scratched or damaged. A suitable solvent together with a soft cloth or soft brush should be used. Please ensure that the tangential aerosol inlet bore is not clogged with fluff from cloth or paper. The generator is reassembled in reverse order.

Note

After using the instrument with salt suspensions, e.g. NaCl or KCl, the instrument must be flashed with distilled water. Therefore please exchange the liquid reservoir and let the generator operate approx. 10 or 15 minutes at a pre-pressure of at least 2 bar. Within the distilled water the salt is flashed out of the tubing and nozzle.

If this procedure was forgotten, it is possible to dismount the dispersion nozzle and clean it within an ultra sonic bath.

Therefore the four screws at the nozzle have to be removed and the dispersion nozzle has to be taken carefully out by pulling. If you reassemble the dispersion nozzle please take care that the part with the opening is facing in the direction of the liquid tubing.

Maintenance by the manufacturer

Maintenance tasks

Maintenance tasks are carried out by the manufacturer. Send the device to the manufacturer.

Address:

Palas GmbH
Greschbachstraße 3 b
76229 Karlsruhe
Germany

6 Troubleshooting

6.1 Nozzle Blockages

If solutions and suspensions are used that contain substances solid at room temperature (e.g. latex particles) or that might crystallize out (e. g. KCl-solutions), the nozzle might become clogged. The generator should therefore be operated with distilled water (with a pure solvent respectively) after every operation, in order to wash out all remaining aerosol substances from the nozzle. If this shall be done, the liquid reservoir must first be emptied and then be re-filled with the solvent (distilled water respectively).

When changing the aerosol substance, the cyclone has to be cleaned and dried carefully. This is most important if you change from hydrophobe to hydrophile liquids, e. g. from DEHS to a KCl-solution. In this case, the reservoir has to be cleaned at first and then the generator itself should be cleaned with Isopropanol. The instrument should be run with Isopropanol for a short time.

The generator must be totally dry when the salt solution is filled in. This state is reached by running the instrument with compressed air for some time. In addition, it must be sure that there are no droplets inside the supply air tube.

If the generator is run with highly concentrated KCl solutions for a long time, there could arise problems concerning the nozzle. They are caused by sedimentations. Ususally, the problems are solved by running the aerosol generator with distilled water for a short time. If this is not enough, the nozzle must be taken out for cleaning purposes. Therefore, the two screws at the nozzle must be screwn off. Then the nozzle can be pulled out. It is important that the O-ring is situated at the correct place when the nozzle is put in again. The O-ring seals the nozzle towards the cyclone.

6.2 Pollution of the aerosol generator

To clean the instrument please firstly removes the material out of the liquid reservoir. After that, please use distillated water to flash the material out of the instrument. Therefore please use at least a pre-pressure of 2 bar. For further information, see the section "Maintenance [▷22]"

For other problems please contact Palas directly if a problem persists or cannot be resolved with the information from the following sections:

Address:

Palas GmbH
Greschbachstraße 3 b
76229 Karlsruhe
Germany

Note

No components may be serviced or replaced in the aerosol generator AGK 2000 without instruction from a Palas employee. In the event of infringement, the customer loses their warranty claims; in addition Palas is not liable for the damage arising.

In case of problems with the aerosol generator AGK 2000, please contact Palas GmbH (Tel.: 0721/96 213-0).

7 Transport, packaging and storage

Package the device securely to exclude damage during transportation.

7.1 Packaging

Please keep the original packaging, so that if it should become necessary (for service or maintenance) you can use this packaging to send the unit back to Palas.

7.2 Storage

If the unit will not be used for a longer period, the instrument should be cleaned. Then the unit should be stored in a dry location.

7.3 Transport

When transporting by hand take care about the weight. For further information, see the section “Technical data [p.27]”

For all other types of transport (such as shipping), the unit must first be cleaned and the instrument should be packed safely for transportation. Please use the original transport packaging.

7.4 Packaging and transportation

Package the device securely to exclude damage during transportation.

Original packaging

For the purpose of shipping, use the original packaging including the protective inner packaging or the original transport case (if applicable).

If you no longer have the original packaging, use packaging that securely protects the device against detrimental influences during transportation.

Detrimental influences during transportation may include temperature, impacts, falling, or vibrations.

8 Disposal

Once aerosol generator *AGK 2000* has reached the end of its lifetime, please contact Palas (0721-96213-0) for proper disposal.

Transport packaging

You can dispose of the transport packaging at collection points.

- Observe the applicable national regulations.

Device and accessories

1. If a battery is installed in the device, make sure that it is properly disposed of.
2. Observe the applicable national regulations.

9 Spare parts list

The following replacement parts are available:

- Dispersing nozzle
- Manometer
- Total filter
- Reservoir glass
- Tubings
- ▶ Only use original spare parts.

10 Technical data

Aerosol substance	KCl-solutions and other solutions and suspensions, such as DEHS, DOP, Paraffin oil and other resin-free oils
Generator material	
Nozzle	POM plastics (delrin)
Cyclone	PP-polypropylene
Seals	• Silicone rubber • Viton • NBR Buna N
Liquid Reservoir	glass
Volume Flow Rate	qprox. 3 - 10 l/min
Concentration	10^6 - 10^7 particles/cm ³ (depends on aerosol)
Generation	24 hrs. non-stop (for pure fluids and KCl-solutions of < 10 % KCl concentration)
Dimensions	H x W x Ø : 400 x 250 x 180 mm
Weight	approx. 3 kg
Pressurized Air	4 bar to 6 bar absolute, 1.5 m ³ /h, dry, oil free and particle free

11 Literature

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